Taller 1, punto 10: Adaline sobre las particiones de Iris

Igual que el punto 4 pero con adaline.m. Los mismos dos problemas: A) setosa contra el resto (separable) B) versicolor contra virginica (no separable) La salida deseada va en [0,1] y se clasifica con umbral 0.5.

```
clear; close all; clc
load iris_splits.mat
if ~exist('informe/figuras', 'dir'), mkdir('informe/figuras'); end
if ~exist('informe/tablas', 'dir'),  mkdir('informe/tablas');  end
```

Parametros

```
op.umbral     = 0.5;
op.salida     = [0 1];
op.error_min  = 1e-3;
op.max_epocas = 500;
op.semilla    = 1;
alphas = [0.001 0.005 0.01 0.05 0.1];

problemas = {'A: setosa vs resto', 'B: versicolor vs virginica'};
```

Entrenamiento

```
filas = {};
curvas = {};
for pr = 1:2
    for k = 1:numel(S)
        if pr == 1
            Xtr = S(k).Xtr;  Xte = S(k).Xte;
            dtr = double(S(k).ytr == 1);  dte = double(S(k).yte == 1);
        else
            mtr = S(k).ytr ~= 1;  mte = S(k).yte ~= 1;
            Xtr = S(k).Xtr(mtr, :);  Xte = S(k).Xte(mte, :);
            dtr = double(S(k).ytr(mtr) == 2);  dte = double(S(k).yte(mte) == 2);
        end
        mn = min(Xtr); mx = max(Xtr);
        Xtr = (Xtr - mn) ./ (mx - mn);
        Xte = (Xte - mn) ./ (mx - mn);

        for a = alphas
            o = op; o.alpha = a;
            [w, b, info] = adaline(Xtr, dtr, o);
            acc_tr = mean(predecir(Xtr, w, b, o.umbral, o.salida) == dtr);
            acc_te = mean(predecir(Xte, w, b, o.umbral, o.salida) == dte);
            etiqueta = sprintf('%d-%d', round(100*S(k).p), round(100*(1-S(k).p)));
            filas(end+1, :) = {problemas{pr}, etiqueta, a, info.epocas,
info.ec_final, ...
                               info.convergio, 100*acc_tr, 100*acc_te}; %#ok<SAGROW>
            if k == 1 && a == 0.01, curvas{pr} = info.ec_por_epoca; end %#ok<SAGROW>
```

```
        end
    end
end
R = cell2table(filas, 'VariableNames', ...
    {'Problema', 'Particion', 'Alpha', 'Epocas', 'EC_final', 'Convergio',
'Exact_train', 'Exact_test'});
disp(R)
```

Problema	Particion	Alpha	Epocas	EC_final	Convergio	Exact_train	Exact_
{'A: setosa vs resto' }	{'60-40'}	0.001	500	0.95925	false	100	1
{'A: setosa vs resto' }	{'60-40'}	0.005	500	0.94532	false	100	1
{'A: setosa vs resto' }	{'60-40'}	0.01	500	0.94854	false	100	1
{'A: setosa vs resto' }	{'60-40'}	0.05	500	0.96619	false	100	1
{'A: setosa vs resto' }	{'60-40'}	0.1	337	1.0079	true	100	1
{'A: setosa vs resto' }	{'70-30'}	0.001	500	1.1164	false	100	1
{'A: setosa vs resto' }	{'70-30'}	0.005	500	1.1078	false	100	1
{'A: setosa vs resto' }	{'70-30'}	0.01	500	1.1114	false	100	1
{'A: setosa vs resto' }	{'70-30'}	0.05	464	1.1272	true	100	1
{'A: setosa vs resto' }	{'70-30'}	0.1	290	1.1792	true	100	1
{'A: setosa vs resto' }	{'80-20'}	0.001	500	1.3227	false	100	1
{'A: setosa vs resto' }	{'80-20'}	0.005	500	1.3045	false	100	1
{'A: setosa vs resto' }	{'80-20'}	0.01	500	1.3028	false	100	1
{'A: setosa vs resto' }	{'80-20'}	0.05	458	1.3071	true	100	1
{'A: setosa vs resto' }	{'80-20'}	0.1	273	1.3664	true	100	1
{'A: setosa vs resto' }	{'90-10'}	0.001	500	1.502	false	100	1
{'A: setosa vs resto' }	{'90-10'}	0.005	500	1.471	false	100	1
{'A: setosa vs resto' }	{'90-10'}	0.01	500	1.4639	false	100	1
{'A: setosa vs resto' }	{'90-10'}	0.05	417	1.474	true	100	1
{'A: setosa vs resto' }	{'90-10'}	0.1	237	1.5381	true	100	1
{'B: versicolor vs virginica'}	{'60-40'}	0.001	500	1.53	false	98.214	93.1
{'B: versicolor vs virginica'}	{'60-40'}	0.005	500	1.3462	false	98.214	93.1
{'B: versicolor vs virginica'}	{'60-40'}	0.01	500	1.3563	false	98.214	93.1
{'B: versicolor vs virginica'}	{'60-40'}	0.05	283	1.4672	true	98.214	93.1
{'B: versicolor vs virginica'}	{'60-40'}	0.1	208	1.6036	true	98.214	95.4
{'B: versicolor vs virginica'}	{'70-30'}	0.001	500	1.7948	false	98.529	90.6
{'B: versicolor vs virginica'}	{'70-30'}	0.005	500	1.6266	false	98.529	90.6
{'B: versicolor vs virginica'}	{'70-30'}	0.01	500	1.6359	false	98.529	90.6
{'B: versicolor vs virginica'}	{'70-30'}	0.05	238	1.7434	true	98.529	90.6
{'B: versicolor vs virginica'}	{'70-30'}	0.1	167	1.8666	true	98.529	90.6
{'B: versicolor vs virginica'}	{'80-20'}	0.001	500	2.4144	false	98.734	95.2
{'B: versicolor vs virginica'}	{'80-20'}	0.005	500	2.2738	false	97.468	95.2
{'B: versicolor vs virginica'}	{'80-20'}	0.01	500	2.2715	false	97.468	95.2
{'B: versicolor vs virginica'}	{'80-20'}	0.05	182	2.4247	true	98.734	95.2
{'B: versicolor vs virginica'}	{'80-20'}	0.1	85	2.6128	true	97.468	90.4
{'B: versicolor vs virginica'}	{'90-10'}	0.001	500	2.6435	false	96.739	1
{'B: versicolor vs virginica'}	{'90-10'}	0.005	500	2.5348	false	97.826	1
{'B: versicolor vs virginica'}	{'90-10'}	0.01	500	2.5382	false	97.826	1
{'B: versicolor vs virginica'}	{'90-10'}	0.05	197	2.7154	true	95.652	1
{'B: versicolor vs virginica'}	{'90-10'}	0.1	97	2.9291	true	96.739	1

```
writetable(R, 'informe/tablas/ada_iris.csv');
```

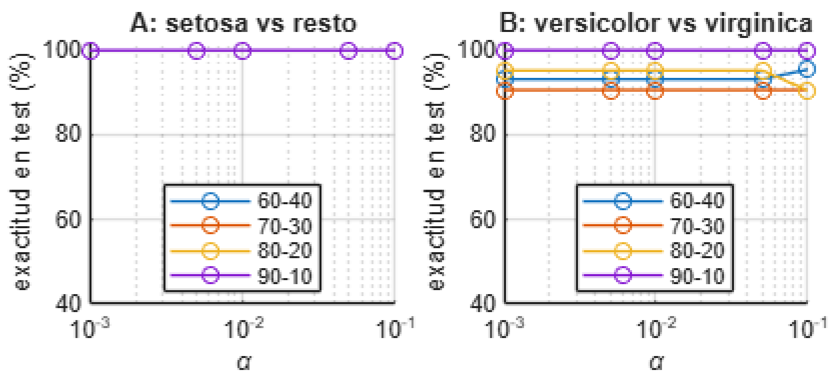
Resumen por problema y particion

```
Res = groupsummary(R, {'Problema', 'Particion'}, 'mean', {'Epocas', 'EC_final',
'Exact_train', 'Exact_test'});
disp(Res)
```

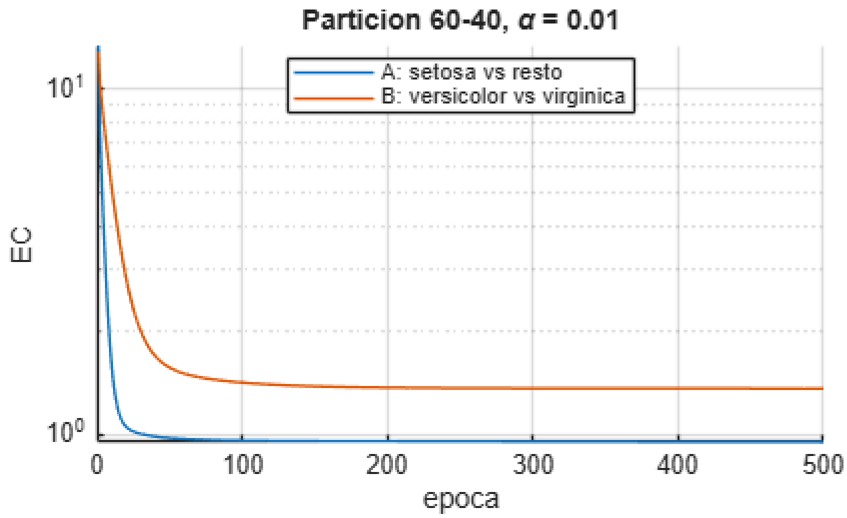
Problema	Particion	GroupCount	mean_Epocas	mean_EC_final	mean_Exact_train
{'A: setosa vs resto' }	{'60-40'}	5	467.4	0.96545	100
{'A: setosa vs resto' }	{'70-30'}	5	450.8	1.1284	100
{'A: setosa vs resto' }	{'80-20'}	5	446.2	1.3207	100
{'A: setosa vs resto' }	{'90-10'}	5	430.8	1.4898	100
{'B: versicolor vs virginica'}	{'60-40'}	5	398.2	1.4607	98.214
{'B: versicolor vs virginica'}	{'70-30'}	5	381	1.7334	98.529
{'B: versicolor vs virginica'}	{'80-20'}	5	353.4	2.3994	97.975
{'B: versicolor vs virginica'}	{'90-10'}	5	358.8	2.6722	96.957

Graficas

```
figure('Position', [100 100 950 380]);
for pr = 1:2
    subplot(1, 2, pr); hold on
    for k = 1:numel(S)
        etiqueta = sprintf('%d-%d', round(100*S(k).p), round(100*(1-S(k).p)));
        m = strcmp(R.Problema, problemas{pr}) & strcmp(R.Particion, etiqueta);
        plot(R.Alpha(m), R.Exact_test(m), '-o', 'DisplayName', etiqueta);
    end
    set(gca, 'XScale', 'log'); grid on; ylim([40 102]);
    xlabel('\alpha'); ylabel('exactitud en test (%)'); title(problemas{pr});
    legend('Location', 'best');
end
guardar_fig('ada_iris');
```




```
figure('Position', [100 100 700 380]); hold on
semilogy(curvas{1}, 'DisplayName', problemas{1});
semilogy(curvas{2}, 'DisplayName', problemas{2});
set(gca, 'YScale', 'log'); grid on
xlabel('epoca'); ylabel('EC'); title('Particion 60-40, \alpha = 0.01');
legend('Location', 'best');
guardar_fig('ada_iris_ec');
```



Particion 60-40, α = 0.01
A: setosa vs resto
B: versicolor vs virginica
EC
10^1
10^0
0
100
200
300
400
500
epoca